

CIRCUIT 2-

Aim:

To design and analyse a Common Source amplifier using 180nm CMOS technology with current source load and to determine:

DC Operating Point Gate Bias Voltages Transistor Widths Voltage Gain Frequency Response
Given:

$V_{DD} = 1.2 \text{ V}$

Power $\leq 0.4 \text{ mW}$

$C_L = 0.5 \text{ pF}$

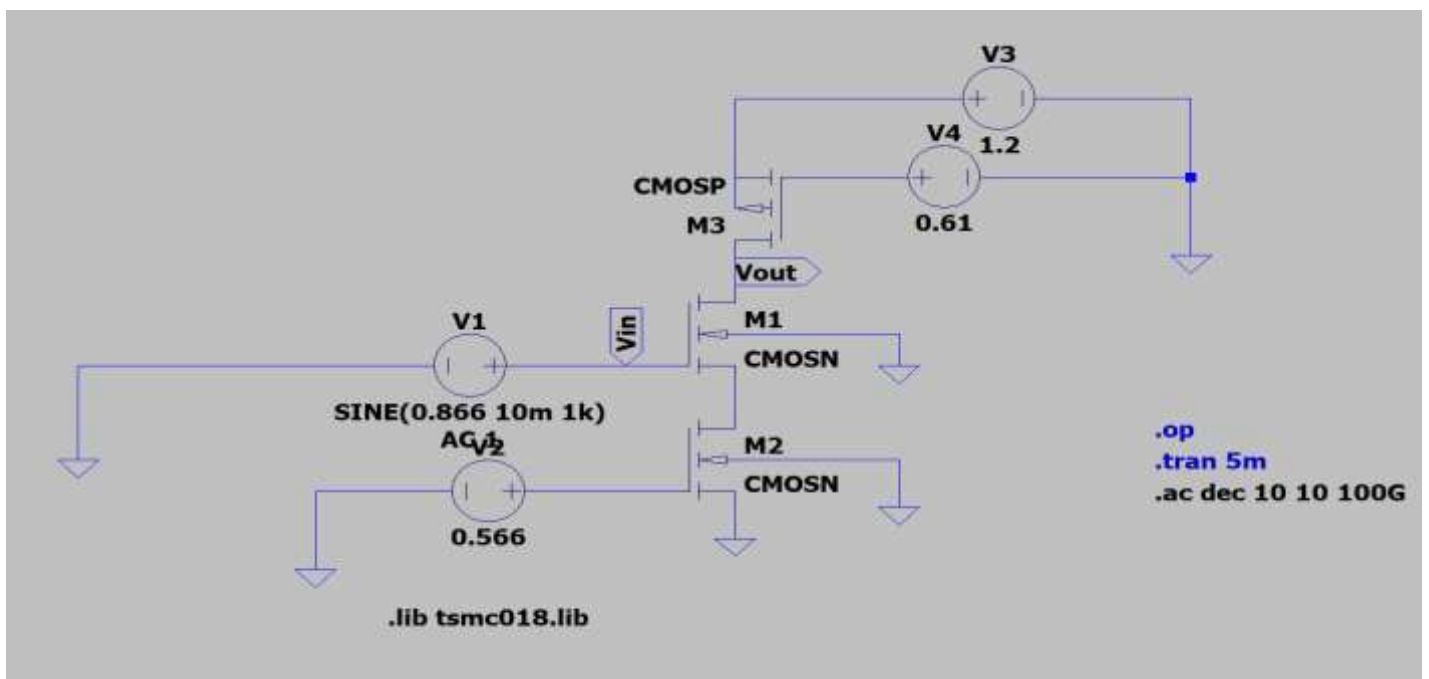
Technology = 180 nm

$L_n = 360 \text{ nm}$

$V_{th(n)} = 0.366 \text{ V}$

$V_{th(p)} = 0.39 \text{ V}$

Circuit Diagram:



Assumed parameters:

$$I_d = 200 \mu A$$

$$V_{ov} = 0.2 V$$

$$\lambda_n = 0.1$$

$$\lambda_p = 0.12$$

calculation:

DC Design Calculations:

Power Verification: $P = V_{DD} \times I_D$

$$P = 1.2 \times 200 \mu A$$

$$P = 0.24 mW$$

Since $0.24 mW < 0.4 mW$

Power condition satisfied.

Gate voltage Calculations: For M1,

$$V_{OV} = V_{GS} - V_{TH}$$

$$V_{GS} = 0.2 + 0.366$$

$$V_{GS} = 0.566 V$$

Assume $V_{S1} = 0.3 V$

$$V_{G1} = V_{GS} + V_{S1}$$

$$V_{G1} = 0.566 + 0.3$$

$$V_{G1} = 0.866 V$$

For M2,

$$V_{S2} = 0 V$$

$$V_{GS2} = 0.566 \text{ V}$$

$$V_{G2} = 0.566 \text{ V}$$

For M3(PMOS),

$$0.2 = V_{SG} - 0.39$$

$$V_{SG} = 0.59 \text{ V}$$

$$V_{G3} = 1.2 - 0.59$$

$$V_{G3} = 0.61 \text{ V}$$

Output Voltage: $V_{out} \approx V_{DD} / 2$

$$V_{out} = 1.2 / 2$$

$$V_{out} = 0.6 \text{ V}$$

Saturation Verification: M3,

$$V_{SD3} = 1.2 - 0.6$$

$$V_{SD3} = 0.6 \text{ V}$$

$$0.6 \geq 0.2 \text{ (satisfied)}$$

M1,

$$V_{DS1} = 0.6 - 0.3$$

$$V_{DS1} = 0.3 \text{ V}$$

$$0.3 \geq 0.2 \text{ (satisfied)}$$

M2,

$$V_{DS2} = 0.3 \text{ V}$$

$$0.3 \geq 0.2 \text{ (satisfied)}$$

Width Calculations: NMOS,

$$I_D = (\mu_n C_{ox} / 2)(W/L)(V_{OV})^2$$

$$200\mu A = (\mu_n C_{ox} / 2)(W/360nm)(0.2)^2$$

$$W_n \approx 7.8 \mu m$$

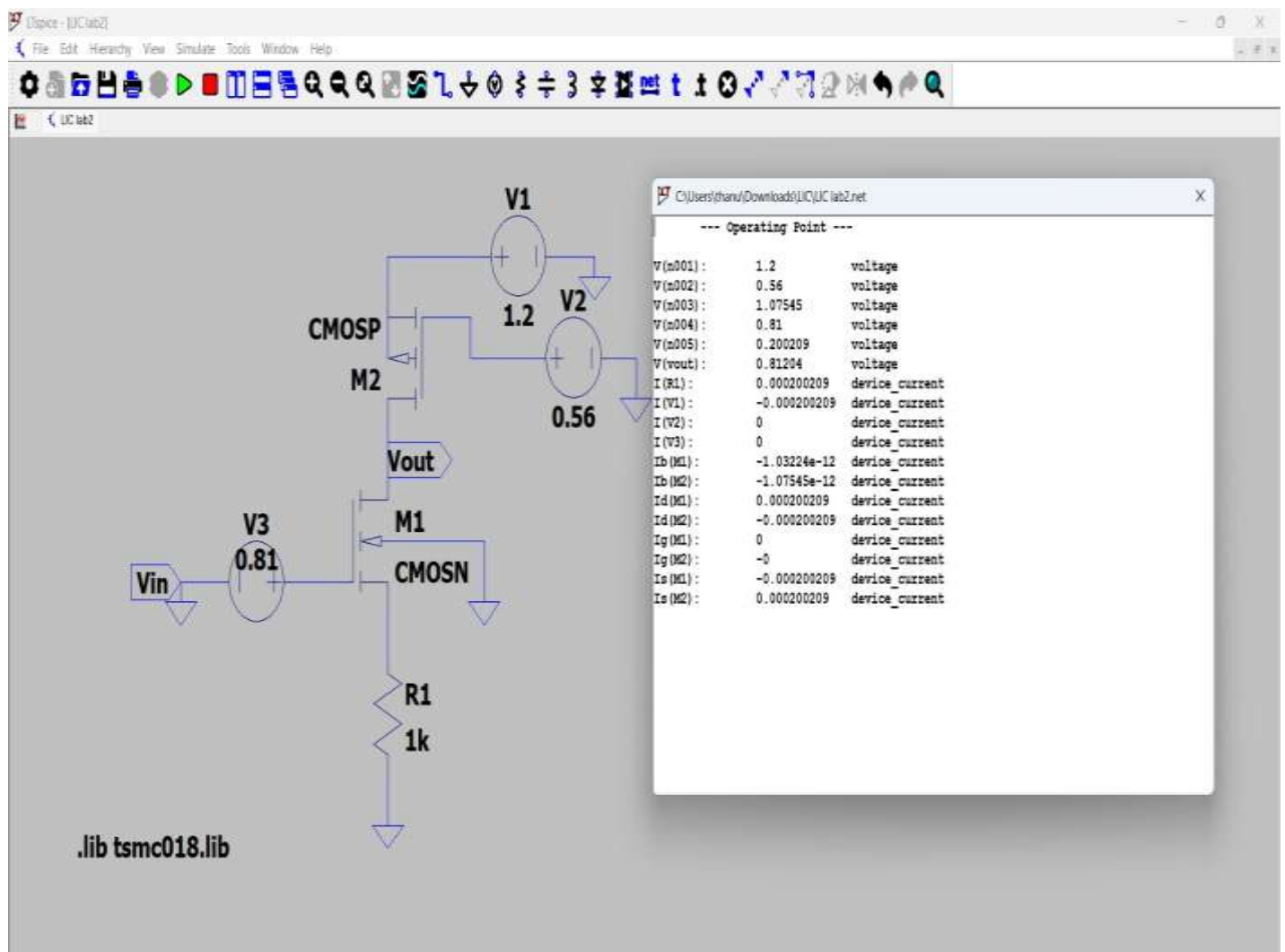
PMOS,

$$200\mu A = (\mu_p C_{ox} / 2)(W/360nm)(0.2)^2$$

$$W_p \approx 18.47 \mu m$$

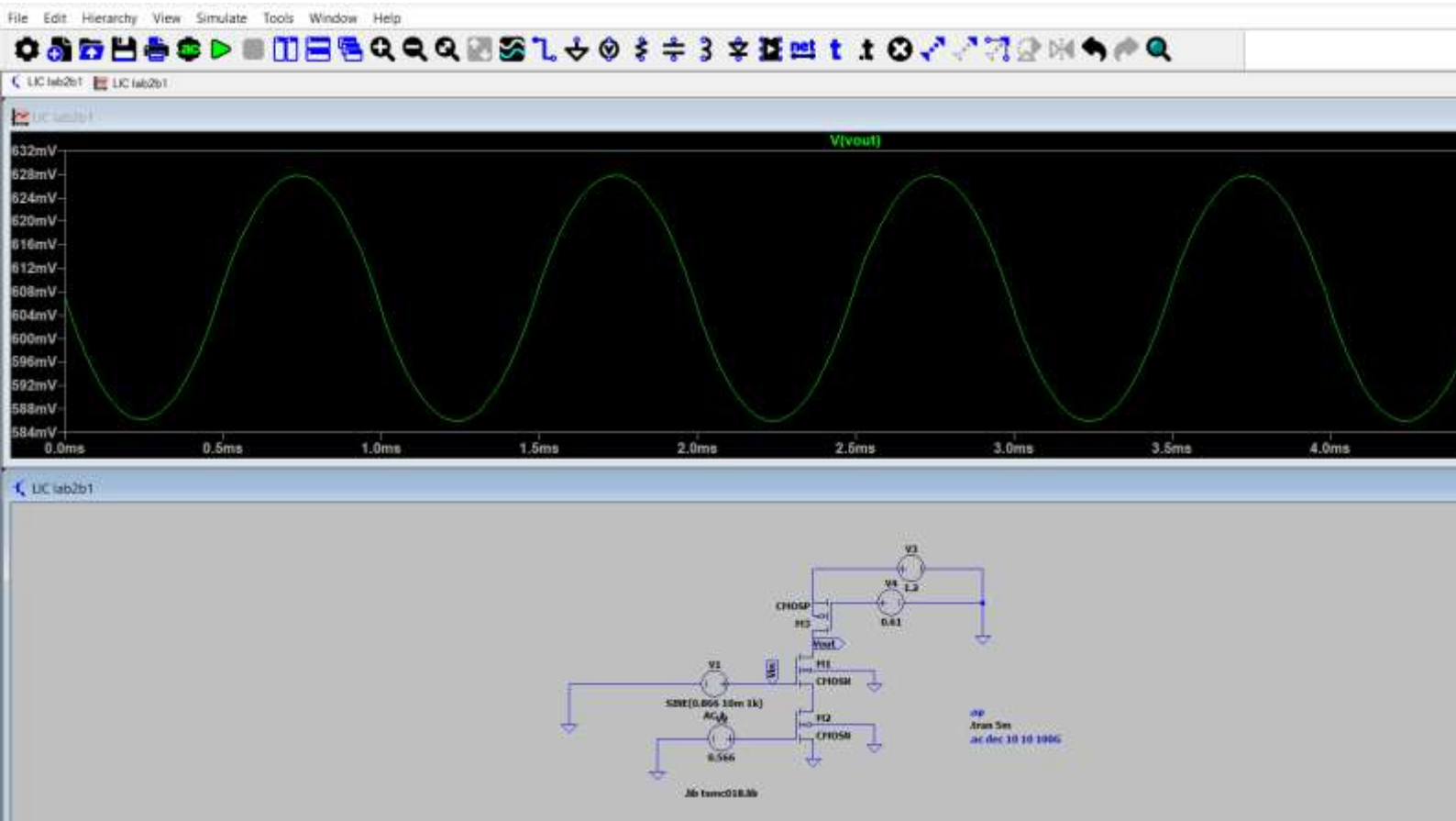
and we should put these values in simulation and we have to do Dc operating for this W value we will get I_d has $50\mu A$, so we should increase the width to increase the current.

DC operating point:



So at $W_n = 34.305\mu\text{m}$ and $W_p = 44.873\mu\text{m}$ the current is $200\mu\text{A}$ and V_{out} is 0.6V biasing is verified.

Transient Analysis:



$$V_{in}(p-p) = 20 \text{ mV}$$

$$V_{out}(p-p) \approx 41.89 \text{ mV}$$

$$A_v \approx 2.094 \text{ V/V}$$

$$A_v(\text{dB}) \approx 6.4 \text{ d}$$

theoretical calculation for gain: $g_m = 2I_D / V_{OV}$

$$g_m = (2 \times 200\mu\text{A}) / 0.2$$

$$g_m = 2 \text{ mS}$$

$$r_{o1} = 1 / (0.1 \times 200\mu\text{A})$$

$$r_{o1} = 50 \text{ k}\Omega$$

$$r_{o3} = 1 / (0.12 \times 200\mu\text{A})$$

$$r_{o3} = 41.66 \text{ k}\Omega$$

$$R_{out} = r_{o1} \parallel r_{o3}$$

$$R_{out} \approx 22.7 \text{ k}\Omega$$

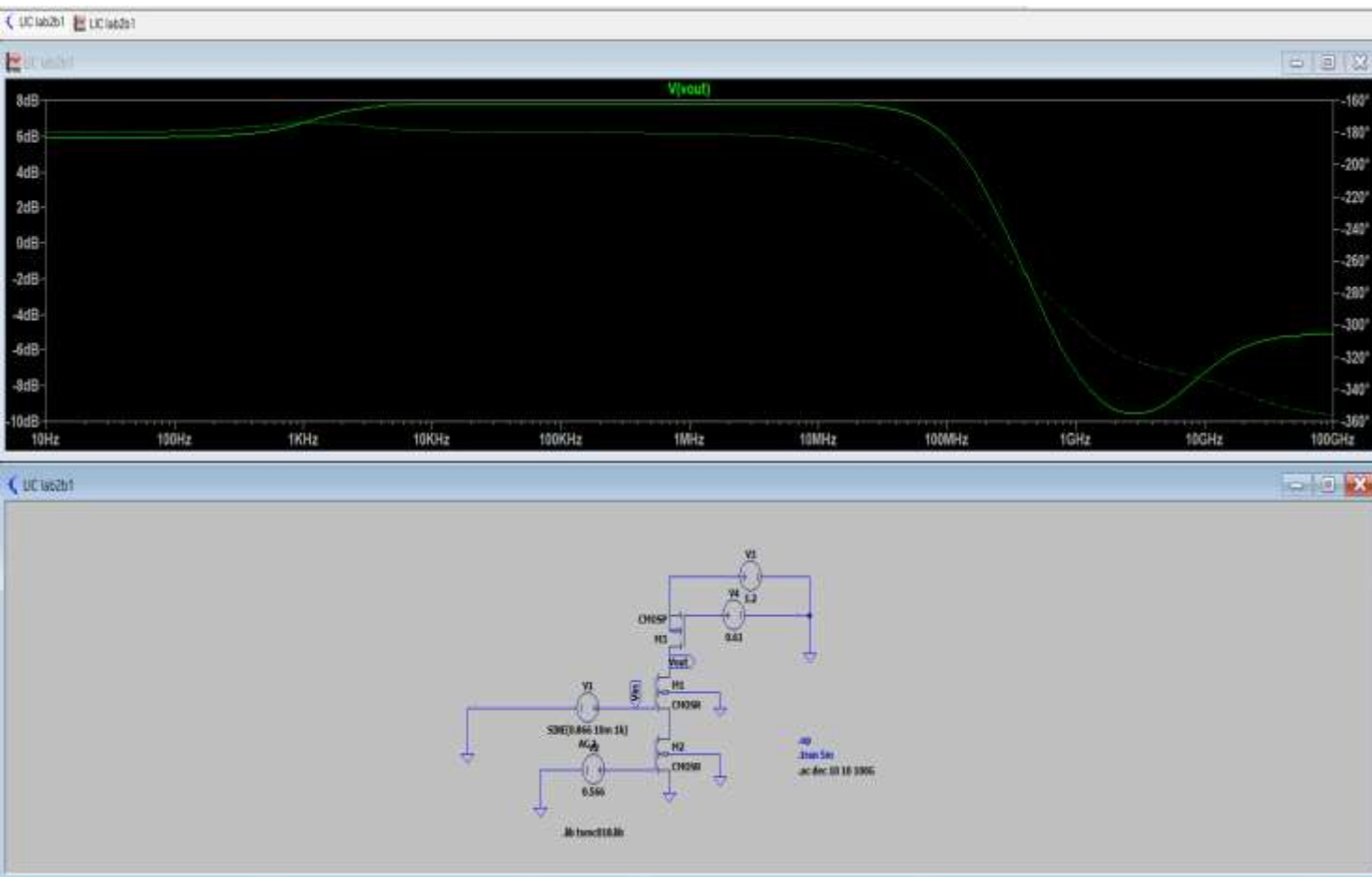
$$\text{Theoretical gain: } A_v = (-g_m(r_{o1} \parallel r_{o3})) / (1 + (g_m r_{o2}))$$

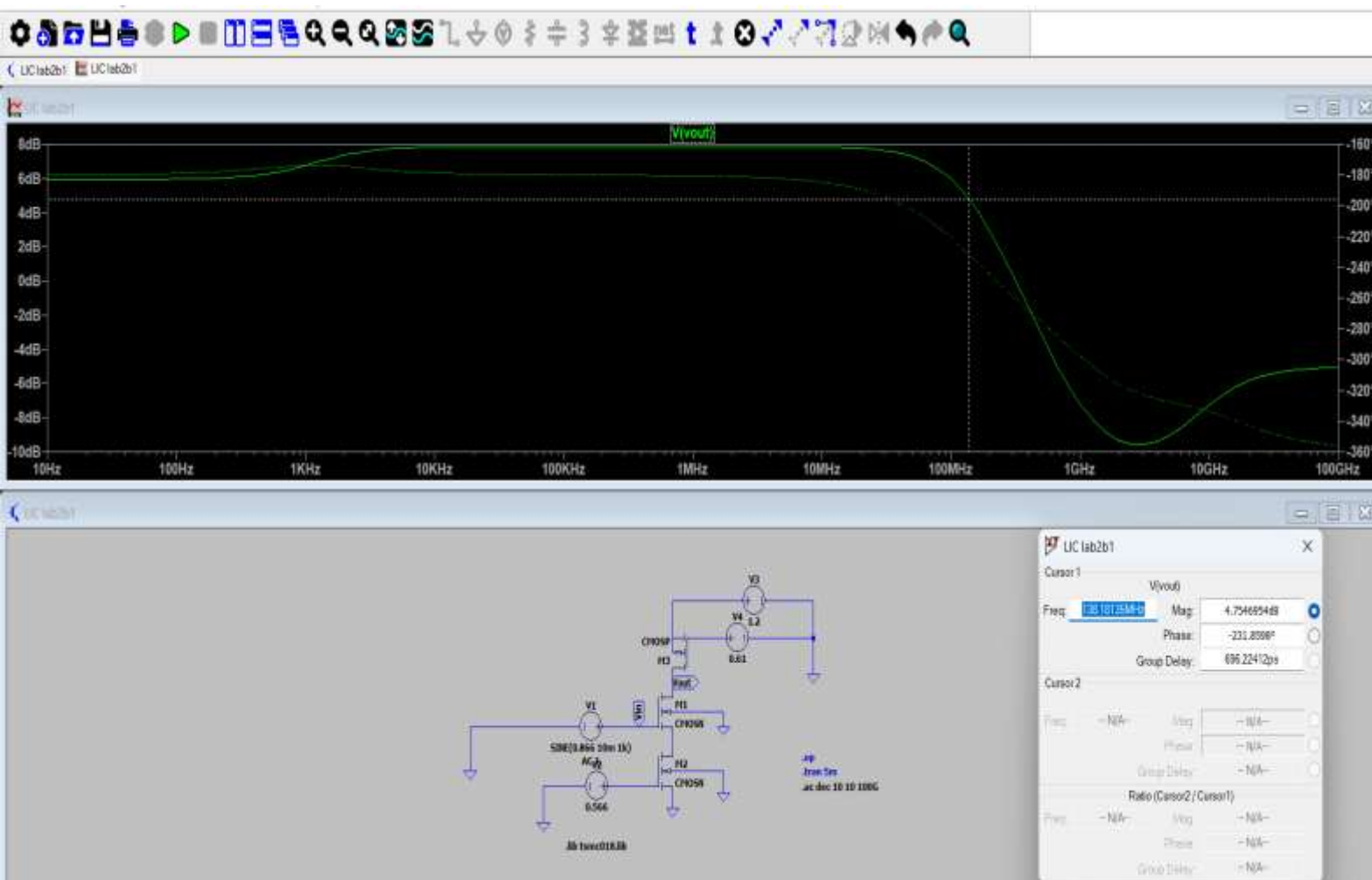
$$A_v \approx -0.45 \text{ V/V}$$

$$A_v(\text{dB}) = 20 \log(0.45)$$

$$A_v(\text{dB}) \approx -6.9357 \text{ dB}$$

AC Analysis:





`.ac dec 100 10 100G`

Midband Gain ≈ 6.4 dB

$f_H = 138.181\text{MHz}$

$f_L = 0\text{Hz}$

BW = 138.151MHz

inference:

All MOSFETs operate in saturation.

Power constraint satisfied.

AC and transient gains match closely.

Result:

CMOS amplifier achieves $\sim 4.75\text{--}16$ dB simulated gain with a bandwidth of ~ 376 MHz, lower than the theoretical 23.74 dB gain due to practical MOSFET non-idealities.